

Comparison Operators: Making Decisions

Comparison operators compare values and return a Boolean result (True or False). They're essential for implementing trading rules:

- == Equal to: if current_price == target_price:
- != Not equal to: if market_status != "Open":
- < Less than: if current_price < stop_loss:
- > Greater than: if rsi > overbought_level:
- <= Less than or equal to: if shares_available <= 0:
- >= Greater than or equal to: if profit >= target_profit:

Here's how a simple trading condition might look:

```
1. # Current market conditions
2. current_price = 142.58
3. moving_average_50 = 140.25
4. moving_average_200 = 135.75
5.
6. # Check for golden cross (short-term MA above long-term MA)
7. golden_cross = moving_average_50 > moving_average_200
8.
9. # Check if price is above both moving averages
10. price_above_averages = (current_price > moving_average_50) and (current_price >
moving_average_200)
11.
12. # Potential buy signal
13. buy_signal = golden_cross and price_above_averages
14.
15. print(buy_signal) # Shows True
```

Logical Operators: Combining Conditions

Logical operators combine multiple conditions:

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